INFLUENCE OF BOUNDARY CONDITIONS ON COMPUTATION OF THE EFFECTIVE THERMAL CONDUCTIVITY OF FOAMS

ME618 TERM PROJECT

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Based on:
Low et al., International Journal of Heat and Mass Transfer, 2020

ABSTRACT

Accurate numerical simulation of the effective thermal conductivity (ETC) of 3D pore-scale foam models requires a judicious choice of boundary conditions, as the computational domains are often smaller than the representative volume element, giving rise to considerable edge effects. Within the finite element homogenization framework, a set of mixed, uniform and periodic boundary conditions are considered. Validity criteria and order relations, demonstrated from entropy-based principles, are numerically verified on random virtual periodic and non-periodic foams. A statistical treatment based on the integral range provides confidence intervals for the estimated ETC.

OVERVIEW

- OBJECTIVE
- THEORETICAL FRAMEWORK
- ELASTICITY ANALOGY
- RVE
- MACROHOMOGENEITY
- MAXIMUM ENTROPY
- GOVERNING EQUATIONS
- BOUNDARY CONDITIONS
- VALIDITY OF MBC
- THEORY IMPLEMENTATION
- FEM WORKFLOW
- EXPECTED RESULT

OBJECTIVE

This project aims to study the influence of boundary conditions on effective thermal conductivity and compare four boundary value problems:

- UTG (Uniform Temperature Gradient)
- UHF (Uniform Heat Flux)
- Periodic Boundary Conditions
- Mixed Boundary Conditions

We attempt to understand the order relations and bias introduced by each BC.

THEORETICAL FRAMEWORK

Heat transfer within engineering parts or structures made of foams typically occur over length scales much greater than the characteristic sizes of the foams' heterogeneities. Applying the principle of scale separation, two distinct scales can be considered.

MICROSCOPIC

Heterogeneous microstructure
Fields: T(x), q(x)

Micromechanics assumes length scale seperation

$$l \ll L$$

where

- l = characteristic microstructure size
- L = macroscopic scale

This allows HOMOGENISATION.

MACROSCOPIC

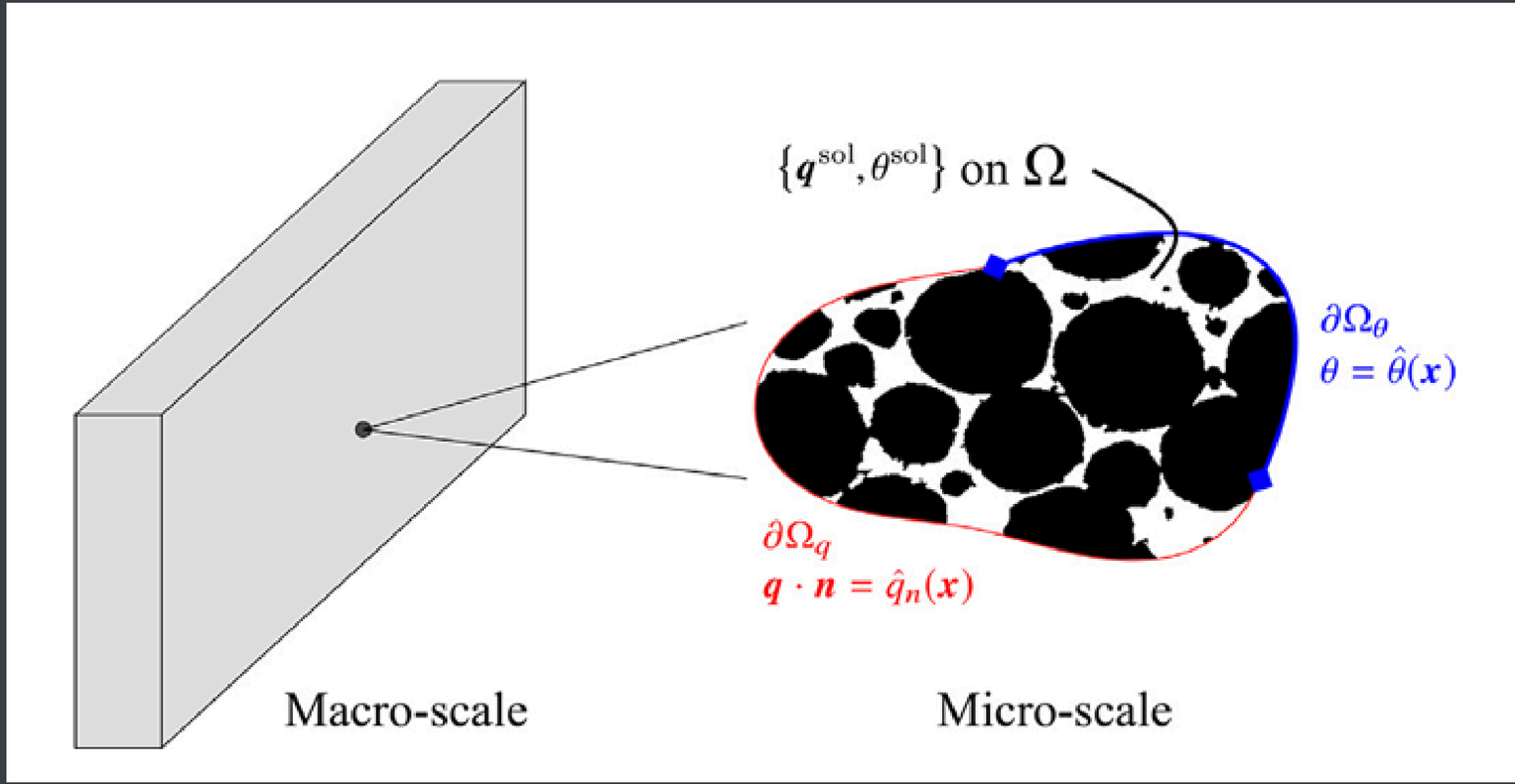
Homogenized behavior

Behavior of the foam may be represented by an effective thermal conductivity (ETC) tensor k eff.

$$\langle q \rangle = -k_{eff} \langle \nabla T \rangle$$

LENGTH SCALE SEPERATION

this is an illustration of the two scale problem for effective visualisation



RVE

A Representative Volume Element is the smallest domain that statistically represents the material microstructure.

Averaging operator:

$$\langle \phi \rangle = \frac{1}{|\Omega|} \int_{\Omega} \phi \, d\Omega$$

Macroscopic quantities:

$\langle q \rangle$, $\langle \nabla T \rangle$

i.e. it is defined such that

$k_{app}(V) \rightarrow k_{eff}$

as

$V \rightarrow V_{rve}$

ANALOGY

Linear Elasticity	Heat Transfer
Displacement (u)	Temperature (T)
Strain ($\varepsilon = \nabla^s u$)	Temperature gradient (∇T)
Stress (σ)	Heat flux (q)
Stiffness tensor (C)	Conductivity tensor (k)
Constitutive law: ($\sigma = C:\varepsilon$)	Fourier law: ($q = -k\nabla T$)
Equilibrium: ($\nabla\cdot\sigma=0$)	Conservation: ($\nabla\cdot q=0$)
Traction: ($t = \sigma\cdot nt$)	Normal heat flux: (q.n)
Elastic energy density: ($\sigma:\varepsilon$)	Thermal dissipation: ($q\cdot\nabla T$)
Hill–Mandel ($\langle\sigma:\varepsilon\rangle=\langle\sigma\rangle:\langle\varepsilon\rangle$)	Macrohomogeneity $\langle q\cdot\nabla T\rangle=\langle q\rangle$
Effective stiffness (C_{eff})	Effective conductivity (k_{eff})

Heat conduction in heterogeneous media is mathematically analogous to linear elasticity, allowing homogenization principles to be directly transferred between the two fields.

MACROHOMOGENEITY CONDITION

Macrohomogeneity condition ensures thermodynamic consistency. The macrohomogeneity condition states that the specific entropy production rate should be conserved between the micro- and macro-scales i.e. energy transfer at the macro scale equals the averaged microscopic energy transfer.

$$\langle q \cdot \nabla T \rangle = \langle q \rangle \cdot \langle \nabla T \rangle = F.G$$

*this is the Simplified Macrohomogeneity Condition for small temperature variations ($T \approx To$) across the boundary. **This condition should be satisfied by the applied boundary conditions.**

MAXIMUM ENTROPY PRINCIPLE

An “admissible” solution for the heat flux field q_adm is a field that satisfies the boundary heat flux conditions, while possibly differing from the “true” solution q_sol elsewhere. The maximum entropy principle states that q_sol maximizes the functional S i.e. is the least biased.

$$\forall \boldsymbol{q} \in \{\boldsymbol{q}^{\mathrm{adm}}\},\ \ \mathrm{S}(\boldsymbol{q}) = \frac{1}{T_0^2}\left(-\int_{\partial\Omega_\theta} (\boldsymbol{q}\cdot\boldsymbol{n})\hat{\theta}\,\mathrm{dA} - \frac{1}{2}\int_{\Omega} \boldsymbol{q}\cdot(\boldsymbol{k}^{-1}\cdot\boldsymbol{q})\mathrm{dV}\right) \leq S(\boldsymbol{q}^{\mathrm{sol}}) \tag{5}$$

where $\hat{q}_n$ is the boundary normal heat flux prescribed on $\partial\Omega_q$.

$$\forall \theta \in \{\theta^{\mathrm{adm}}\},\ \ \mathrm{S_c}(\theta) = \frac{1}{T_0^2}\left(-\int_{\partial\Omega_q} \hat{q}_n\theta\,\mathrm{dA} - \frac{1}{2}\int_{\Omega} (\boldsymbol{k}\cdot\nabla\theta)\cdot\nabla\theta\,\mathrm{dV}\right) \leq S_c(\theta^{\mathrm{sol}}) \tag{6}$$

where $\hat{\theta}$ is the boundary temperature prescribed on $\partial\Omega_\theta$.

*this is the complementary maximum entropy statement

GOVERNING EQUATIONS

Macro-Scale Fourier's Law

$$\boldsymbol{F} = -\boldsymbol{k}_{\text{eff}} \cdot \boldsymbol{G} \text{ with } \nabla \cdot \boldsymbol{F} = 0$$

where:

- F = macroscopic heat flux
- G = macroscopic temperature gradient

Micro-Scale Fourier's Law

$$\boldsymbol{q} = -\boldsymbol{k} \cdot \nabla\theta \text{ with } \nabla \cdot \boldsymbol{q} = 0$$

where:

- q = micro-scale heat flux vector
- θ = relative temperature

Scale Transition

$$\boldsymbol{F} = -\boldsymbol{k}_{\text{app}} \cdot \boldsymbol{G} \equiv \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = -\begin{bmatrix} k_{11}^{\text{app}} & k_{12}^{\text{app}} & k_{13}^{\text{app}} \\ k_{21}^{\text{app}} & k_{22}^{\text{app}} & k_{23}^{\text{app}} \\ k_{31}^{\text{app}} & k_{32}^{\text{app}} & k_{33}^{\text{app}} \end{bmatrix} \begin{bmatrix} G_1 \\ G_2 \\ G_3 \end{bmatrix}$$

where:

- F ≡ <q> and G ≡ <∇ θ> are averaged quantities

$$\langle \bullet \rangle = \frac{1}{V} \int_{\Omega} \bullet \, \mathrm{d}V$$

Steady State Heat Conduction

$$\nabla \cdot (k \nabla \theta) = 0$$

it ensures the conservation of energy and a physically consistent field solution

BOUNDARY CONDITIONS

The apparent thermal conductivity depends on the choice of boundary conditions, as they define different admissible temperature and heat flux fields over the RVE.

Boundary Condition	Mathematical Definition	Macro Quantity Imposed	Key Property
Uniform Heat Flux (UHF)	$\forall x \in \partial\Omega,\ q \cdot n = Fo \cdot n$	$\langle q \rangle = Fo$	Automatically satisfies macrohomogeneity
Uniform Temperature Gradient (UTG)	$\forall x \in \partial\Omega,\ \theta = Go \cdot x$	$\langle \nabla\theta \rangle = Go$	Automatically satisfies macrohomogeneity
Periodic (PBC)*	$\theta(x+) = \theta(x-) + Go \cdot (x+ - x-)$	Imposes fluctuation periodicity	Ensures anti-periodic normal flux, Macrohomogeneity
Mixed BC (MBC)	Combination of θ and $q \cdot n$	Partially constrained fields	Does **not always** satisfy macrohomogeneity

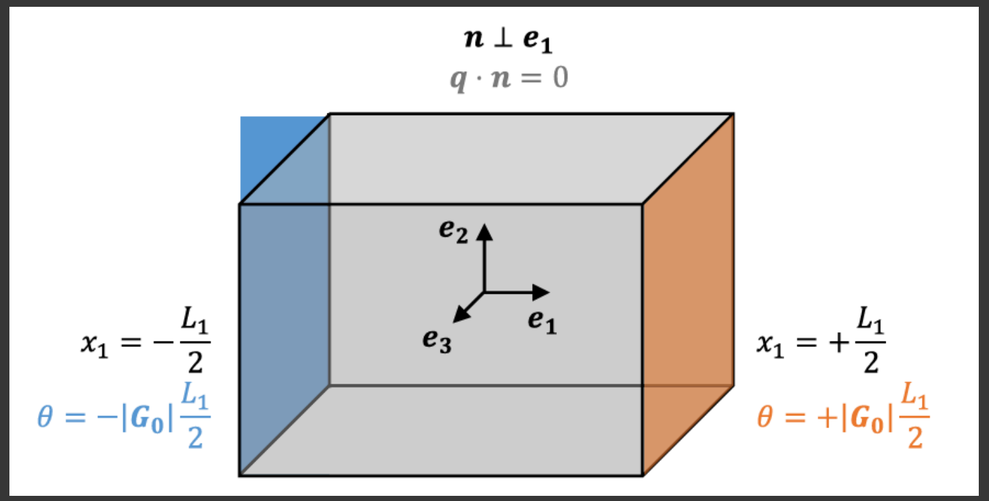
*Periodic boundary conditions are typically implemented on cuboidal RVEs

VALIDITY OF MBC

- Unlike uniform and periodic boundary conditions, mixed boundary conditions (MBC) do not generally satisfy the macro- homogenity condition.

- If violated → inconsistent energy transfer across scales → not always valid for homogenization

- **Criteria for validity : F2G2 + F3G3 = 0**



THEORETICAL IMPLEMENTATION

DEFINE DOMAIN

- Consider a heterogeneous domain Ω
- Domain represents a finite approximation of the RVE

SOLVE LOCALLY

- Apply governing PDE
- Apply Boundary Conditions
- Obtain Temperature field $\theta(x)$ and Heat flux field $q(x)$

HOMOGENISE

- Compute Macroscopic Quantities $G = \langle \nabla\theta \rangle$, $F = \langle q \rangle$
- Determine Apparent Conductivity $F = -k_{app} \cdot G$
- Convergence $k_{app} \to k_{eff}$ as domain $\to$ RVE

For a given finite domain, boundary conditions define admissible temperature and heat flux fields. The steady-state heat conduction equation uniquely determines the physical solution among these admissible fields. THIS SOLUTION ALSO SATISFIES THE MAX ENTROPY PRINCIPLE. Averaging these fields yields macroscopic quantities, from which an apparent conductivity is obtained. This apparent conductivity depends on boundary conditions but converges to the effective conductivity as the domain approaches an RVE.

FEM WORKFLOW

GEOMETRY+ MESHING

- 2D domain (L1=L2=L)
- Periodic
- Non-Periodic

ASSIGN MATERIAL PROPERTIES

- Solid Phase conductivity
- Pore Phase conductivity

APPLY BOUNDARY CONDITIONS

- UTG
- UHF
- PBC
- MBC

RUN SOLVER

- 2D steady-state heat conduction
- Implemented in ABAQUS
- Krylov-based iterative solver

POST PROCESSING

- Output Fields
- Temperature field $\theta(x)$, Heat flux field $q(x)$
- Averaging $G=\langle \nabla\theta \rangle$, $F=\langle q \rangle$
- Compute Conductivity $F=-kapp \cdot G$

VALIDATION

- Validity (Macrohomogeneity)
- Accuracy
- Precision
- Diagonalizability

EXPECTED RESULTS

ADMISSIBLE FIELDS

- The choice of boundary conditions determines the admissible temperature and heat flux fields, which in turn leads to bounded estimates of the apparent thermal conductivity.
- Any heat flux q is an admissible solution heat flux field to the UTG problem, since no boundary heat fluxes are prescribed ($\partial q = \varnothing$).
- Conversely, any temperature θ is an admissible solution temperature field to the UHF problem since no boundary temperatures are defined ($\partial \theta = \varnothing$).

KNOWN ORDER RELATIONS

Assuming that there exists a cubioidal domain that, when subjected to the PBC, gives zero normal heat fluxes on the lateral faces.

For sufficiently large volumes:

$$k_{\text{app}}^{\text{UHF}} \leq k_{\text{eff}} \leq k_{\text{app}}^{\text{UTG}}$$

Comparison between mixed and periodic boundary conditions:

$$k_{\text{app}}^{\text{PBC}} \leq k_{\text{app}}^{\text{MBC}}$$

Combined order relation:

$$k_{\text{app}}^{\text{UHF}} \leq k_{\text{app}}^{\text{PBC}} \leq k_{\text{app}}^{\text{MBC}} \leq k_{\text{app}}^{\text{UTG}}$$

THANK YOU